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Weighted Finite State TransducersWeighted finite state automaton that transduces an input sequence to an output sequence (Mohri et al 2008)Weighted finite state transducersStates connected by transitions. Each transition has input label output label weight.Weights use the log semi-ring or tropical semi-red - with operations that correspond to multiplication and addition of probabilities.There is a single start state. Any state can optionally be a final state (with a weight) or a new state.Used by Kaldi

Springer Handbook on Speech Processing and Speech Communication Professionally Weighted Finish State Acceptors is 0.5.5 better, 0.6.7. Are/0.5 data/0,0.66 worse/1.3 is/1 using/1 intuition/0-0.33(a) d/2 d/3 d/1 d3 d2 d3(b) d1 d4 d3 dx/1 ae/1 dx/2(c) d2(d3 d3) d3/1dx/1 dst/1 st/1 (d1 d2d3) dst/2Figure 1: Weighted finite-state acceptor examples. By convention, the states are represented by circ marked with their unique number. The initial state is represented by a bold circle, the final state by circles. The label l and weight w of a transition are marked on the corresponding directed arc by l/w explicitly shown, the final weight wof a final state f is marked by f/w. Since p ciation transducers are (quite similar to a weighted acceptor except that it has an input label, an output label and a weight on each the output label, it is possible to combine the p ciate transducers for more than one word

Weighted, Finite State Transducers, ImprovisedSpringer Handbook on Speech Processing and Speech Communication Package Acceptor is 0.0.5 better, 0.5.7. Are/0.5 data/0.66 worse/0.3 is/1 using/1 intuition/0.33 (a) d/1 ey/0.5 t/0.3 ax/1 ae/0.5 dx/0.7 (b) Transducer d1 d2 d3 is/0.5 better:better/0.7. Using:using/1 data:data/0.66 are:are/0 .5 is:is/1 intuition:intuition/1(c) worse:worse/ 0.3 is: is/1 (d) intuition/1t:tuition:0.33 t:ε/0,0.3 ey:ε-0.5 ax: ε/1 dx:ε'0.7 is:δ/1Figure 1: Weighted finite-state acceptor examples. By convention, the states are represented by circ marked with their unique number. The initial state is represented by a bold circle, the final state by circles. The label l and weight w of a transition are marked on the corresponding directed arc by l/w explicitly shown, the final weight wof a final state f is marked by f/w. The label l is marked on a directed arc.ae:ε/0.5 d:data/1 d:dew/1 uw:ε-1 it i il T. t. t. t. t. l. Ht d d h d d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d h d T th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t th t It. H. The t, l, b, l. It. I ibl t bi bi bi.

Using/1 intuition/0.33 is/1 worse/0.3 Weighted Finite State Transducers Springer Handbook on Speech Processing and Speech Communication (a) Acceptor d/1 ey/0.5 t/0.3 ax/1 is:is/0.5 ae/0.5 dx/0.7 better:better/0. (b) using:using/1 data:data/0.66 are:are/0.5 d1 d2 d3 worse:worse/0 is:is/1 intuition By convention, the states are marked with their unique number. The initial state is represented by a bold circle in circles. The label l and weight w of a transition are marked on the corresponding d label. As shown, the final weight of a final state f is marked by f/w. A weighted acceptor is quite similar to a weighted input/output (quite similar to an input label, an output label and a weight on each of its transitions. The examples in Figure 2 encode (a superset of) the information in the WFSAs of Fig- the output label, it is possible for transducers to lose word identity. Sim the form given in Figure 1.Figure 2: Weighted finite-state transducer examples. These are similar to the weighted acc except output labels are introduced on each transition. The input label i, the output label o. A transition are marked on the corresponding directed arc by i.e. dot or dot.

WFST Algorithms for Transducer-Composition Combine transducers T1 and T2 into a single transducer acting as if the output of T1 was passed into T2. Determinisation Ensure that each state has no more than a single output transition for a given input labelMinimisation Transforms a transducer with the fewest possible states and transitions for each input labelWeight pushing Push the weights towards the front of the path

The HMM as a WFST ax_1 ax_2 ax_3 ax_4 ax_5 ax_6 ax_7 ax_8 ax_9 ax_10 ax_11 ax_12 ax_13 ax_14: ax_16 ax_17 ax_18:

Applying WFSTs to speech recognition Represent the following components as WFSTs transducer input sequence output sequence G word-level grammar words L pronunciation lexicon phones C context-dependent CD phones H HMM HMM states Composing L and G results in a transducer that maps a phone sequence to a word sequence H C L G results in a transducer that maps from HMM states to a word sequence H C L G results

Grammar - unigram the:the/8.1333 piper:piper/7.4401 pickled:pickled/7.4401 peppers:peppers/7.4401 peck:peck/7.4401 of:7.4401 a:a/8.1333 0/0.0047087.

Grammar - bigram the:the/0.22 the:the/1.6 picked:picked/0.5 peck:peck/1.38 4/1.6 peck:peck/1.7 piper:piper/0.4 picked:picked/0.4 1/0.6 a:a/1.65 6/1.1 peter:peter/1.1 a:a/1.09 peter:peter/5.81 10/1.1 peter:peter/1.1 a:a/1.09 peter:peter/5.81

Bigram with back-off peter:peter/5.8111 the:the/0.22764 the: the/1.6094 peck:peck/2.8255 the:a/1,10/0,11/2,10-10/12.5,11,11 and11/12,11.5.5

peck,peck,1.3863 of:of/ 5.118 the:their/0-0.19633 a:a-1.6523 The FOR-0.99599525, piper.peripiper.8.37.37376464.3.4.6.1.0.2.1-3.3-3-4.7-7.7449 of For-op0.11.13.2-3,3.1,3-7,4-4,7-4-7-5-5.5.3,4.5-6-5,5-9,746-6.5,6-6,8-8,7 Peter:peter/0.69315 peppers:peppers/8.3764 a:a/1.8546 piper:piper/0.40547 peck:peck/7.6833 peppers:peppers/0.40547 0/0.941 #0:eps>/1.6062 pickled:pickled/8.3764 peter:peter/8.3764 2/0.60614 picked:picked/0.40547 a:a/1.8566 #0 Ance ata-1.1.0.0-0.09861.986.086.2.9-86.9.

Weight pushing piper:piper/4 peter:peter/0.5 picked:picked/10 the:the/1 peck:peck/1.5 peck/12

Weight-pushed version piper:piper peter:peter/4.5 picked:picked/6 the:the/2 peck:peck/13

Lexicon, 'Laughingly'16.1.2.1,1.0,2.2,1,3.0.3,2,4.4,4,6.3.1',1.3 p.s. httptopslight, httptopeter, and httptenslight, endpoint (3.4).

Determinization - det(L) 13 er:eps> p:eps> iy:eps> dh:the k:peck ay:piper 18 z:eps> 14 eh:eps> 10 p:peppers ih:eps> 19 l:eps> 20 d:eps> 15 ah:pickled

Minimization - min(det(L)) I.e.:1.1.2.3.1,1.3,2.2,3.0,1,2,2-eps-1.4,1x1.0.1x2.5,2x1,3,1-piper,4x2,4-pipher,3x3,3d4,4,3-p3,4.0 (2.0-p4.5) This is a growing sector in which many new jobs are being created. The Commission feels this could not be justified on the

Composition piper:piper/4 peter:peter/0.5 picked:picked/10 the:the/1 peck:peck/1 peter/1.17

L G 13 ay:eps> 15 p:eps> 10 p:piper/4 er:eps> t:eps> p:peter/0.5 p:picked/10 11 14 ih:eps> 16 k:eps> 12 ah:eps> p:peck/1 eh:eps> 18

Min(det(L ■G)) K.O.P.L.T.S. P.T., httpteps.p.1.2.1, httptepeter.4.3.2-3.3-3-2.3 httptpeter.3, httpenslight.33.1-2-2 httptptenslight-displayed httptenscreen.4-4.

Context-dependency ey/uw:uw ey/ey:uw uw/uw:uw ey/t:t eps>/ey:ey t/t:t uw/t:t t/ey:ey t/uw:uw eps>/t:t eps>/uw:uw 20.

Eps>/iy/eps>:eps> eps>/iy/p:p p/iy/eps>:eps>/iy/iy:iy/iy/eps>:eps> iy/iy/iy:iy p/iy/iy:iy eps>:iy p/iy/

C ■, ■-G-1 biphones, ah-p3-2-3-1-2P3-3P2-2p3p3P1-4P3P4-4p2-1P3D4-1p3d4-2d4 p4-3p2p4p3a3p1-3b3-4b3p6p3c3p4d4tp3p0p3tp4p6-4dh2p6d4P6p6P3p

HMM transducer, H1.1.0.1,1.2.11,2.2,1,3.4,2,4.3.1-3.2-3,2-4,3-4.4-4-5.3,1-5,3,4,5,6,6.4.0,5.5,5-7,6-7.5-6,7,7-6.5.0-8,7.6,8.5(2.5)1.5

Decoding using WFSTs Combining the transducers gives an overall HMM structure for the ASR system - but minimisation and determination operations on the WFSTs means it is much smaller than naively combining the HMMs. Ors But it is important in which order the algorithms are combined otherwise the transducers may blow-up. A standard approach is to determine and minimize after each composition. Bare minimum makeup with shades of some glow on those sculpted cheeks and tresses tied in a messy ponytail, the actor looked like a princess. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24. 24

Reading the text of al-Mohri et al (2008). Imize speech recognition with weighted finite-state transducers. In Springer Handbook of Speech Processing, pp. 559-584. Springer. Http://www.cs.nyu.edu/~mohri/pub/hbka.pdf.html.WFSTs in Kaldi. Http://danielpovey.com/files/Lecture4.pdf/index.php?id=25